Beyond Loss Curves: Thermodynamics of Pretraining

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Abstract

Training a frontier language model costs hundreds of millions of dollars, yet practitioners monitor this investment through a single, impoverished signal: the training loss. We show that loss curves are a fundamentally incomplete instrument for understanding pretraining dynamics, analogous to monitoring a chemical reaction by temperature alone. We propose a thermodynamic framework that equips pretraining with a full state-variable panel: spectral free energy, spectral entropy, a crystallization order parameter, and thermodynamic efficiency. Measuring these quantities across 4,000+ checkpoints from OLMo models spanning 190M to 13B parameters, we establish four results: (1) pretraining dynamics satisfy a modified thermodynamic state equation that generalizes the ideal gas law recently verified at small scale, with finite-size corrections that vanish as $N^{-1/3}$; (2) Warmup-Stable-Decay traces a more thermodynamically efficient trajectory than cosine decay, with 23–37% lower cumulative entropy production, providing the first physics-grounded explanation for its empirical superiority; (3) mid-training exhibits glass-like relaxation: spectral entropy follows the Kohlrausch-Williams-Watts stretched exponential with $\beta \approx 0.6$, yielding a principled estimate of optimal mid-training duration; (4) a Gaussian decay schedule derived from the minimum entropy production principle outperforms WSD by 2.1% in final loss at matched compute. These results reframe pretraining as a thermodynamic process where the central question shifts from "has loss converged?" to "is the system still producing useful work?"

1 Introduction

Pretraining a state-of-the-art language model is among the most expensive computational endeavors in science and industry. A single frontier run consumes $100–500M in compute, occupies thousands of GPUs for 3–9 months, and processes trillions of tokens of text [DeepSeek-AI, 2024, Meta AI, 2025, OLMo Team et al., 2025]. Throughout this process, practitioners rely on one real-time diagnostic: the training loss curve. This is the equivalent of piloting a $500M aircraft with nothing but an altimeter.

The poverty of loss-only monitoring is not a minor inconvenience; it is a structural blind spot. Loss correlates with final downstream performance at $r < 0.40$ [OLMo Team et al., 2025]—worse than a coin flip for predicting which checkpoint to deploy. MoE routing pattern consistency predicts performance at $r > 0.97$ [Doshi et al., 2025], yet is rarely tracked. Overtraining, now standard practice to reduce inference cost [Gadre et al., 2024], makes models *harder* to fine-tune through a mechanism no existing theory explains [Springer et al., 2025]. The Warmup-Stable-Decay (WSD) learning rate schedule has empirically displaced cosine decay across the industry [Wen et al., 2025, OLMo Team et al., 2025, DeepSeek-AI, 2024], but the only explanation offered is a geometrical conjecture about "river-valley" landscapes [Wen et al., 2025]—descriptive, not predictive. Mid-training duration, arguably the highest-leverage hyperparameter in the multi-stage pipeline, is chosen by grid search over proxy models [OLMo Team et al., 2025, Grattafiori et al., 2024].

Meanwhile, a convergent body of results from statistical physics has established that neural network training is, in a precise mathematical sense, a **thermodynamic process**:

- SGD minimizes *free energy* $\mathcal{F} = \mathcal{U} - \mathcal{T}\mathcal{S}$, not loss alone, where $\mathcal{T}$ is an effective temperature determined by the learning rate and gradient noise [Sadrtdinov et al., 2025a].
- The fluctuation-dissipation relation identifies learning rate as the system temperature [Liu et al., 2025a,b].
- Scale-invariant networks satisfy the ideal gas law $\mathcal{P}\mathcal{V} = Nk\mathcal{T}$, with weight decay as pressure and parameter norm as volume [Sadrtdinov et al., 2025b].
- Grokking—delayed generalization—is a glass relaxation process, not a phase transition [Zhang et al., 2025].
- An algorithm-independent lower bound on entropy production constrains how fast any system can learn: the Epistemic Speed Limit [Okanohara, 2026].
- Deep networks exhibit scaling collapse under renormalization, a hallmark of universality [Qiu et al., 2025].

Yet every one of these results was established at toy scale: GPT-2-small or smaller [Liu et al., 2025a], synthetic arithmetic tasks [Zhang et al., 2025], CIFAR classifiers [Sadrtdinov et al., 2025a,b]. The thermodynamic lens has never been turned on real, large-scale, multi-stage pretraining.

This paper fills that gap. We define operationally measurable thermodynamic state variables for language model pretraining (§3), compute them across 4,000+ OLMo checkpoints from 190M to 13B parameters (§4), and derive four results that change how pretraining should be understood and practiced:

1. **A thermodynamic state equation at scale** (§4.2). Pretraining dynamics satisfy a modified ideal gas law with finite-size corrections, validated across four model scales on modern architectures (RMSNorm, AdamW, RoPE). The state equation predicts training trajectory shapes and identifies three distinct thermodynamic regimes: ideal gas (early training), liquid (stable phase), and solid/glass (post-decay).
2. **A physics-grounded explanation of why WSD outperforms cosine** (§4.3). We measure the entropy production rate—the thermodynamic measure of irreversibility—for both schedules and find that WSD produces 23–37% less cumulative entropy. The isothermal stable phase functions as a quasi-equilibrium exploration period; the subsequent decay is a controlled quench. Cosine's continuous cooling forces the system through non-equilibrium intermediate states that dissipate compute irreversibly.
3. **Mid-training as glass-to-crystal annealing** (§4.4). At the data-switching point, spectral entropy drops sharply and follows the Kohlrausch-Williams-Watts (KWW) stretched exponential $\phi(t) = \exp[-(t/\tau)^{\beta}]$ with $\beta \approx 0.6$, the quantitative signature of glassy relaxation [Williams and Watts, 1970]. The characteristic time τ provides a principled estimate of optimal mid-training duration: $\sim 3\tau$ tokens for 95% relaxation. The crystallization order parameter ψ correlates with downstream benchmark performance at $r > 0.92$, far exceeding loss ($r < 0.40$).
4. **A better schedule from first principles** (§4.6). We derive a Gaussian decay schedule from the minimum entropy production principle [Okanohara, 2026, Liu et al., 2025a] and validate it on proxy experiments at 190M and 1B scale, achieving 2.1% lower final loss than WSD at matched compute.

Together, these results establish that the right mental model for pretraining is not optimization but thermodynamics. The question that should guide every training decision is not "has loss converged?" but "is the system still doing useful thermodynamic work, or is it wasting compute on irreversible dissipation?" We release ThermoSched, a library for computing thermodynamic state variables from checkpoints and deriving optimal schedules.

2 Related Work

Statistical physics of neural networks. The application of statistical mechanics to learning systems has a long history [Engel and Van den Broeck, 2001], but recent work has established *quantitative* correspondences rather than loose analogies. Sadrtdinov et al. [2025a] proved that SGD with finite learning rate minimizes free energy $\mathcal{F} = \mathcal{U} - \mathcal{T}\mathcal{S}$, not loss alone, and Sadrtdinov et al. [2025b] showed that scale-invariant networks with weight decay exactly satisfy the ideal gas law $\mathcal{P}\mathcal{V} = Nk\mathcal{T}$. Liu et al. [2025a] derived three thermodynamic laws for LLM training and showed the optimal schedule corresponds to a quasi-static process, though experiments were limited to GPT-2-small (124M). Liu et al. [2025b] established entropic force theory, proving that SGD stochasticity systematically breaks continuous parameter symmetries. Okanohara [2026] proved the Epistemic Speed Limit—an algorithm-independent lower bound on entropy production for any learning process, analogous to Landauer's principle. Seroussi et al. [2023] derived training speed limits via Wasserstein-2 distance bounds. All of these results were verified only at small scale. Our work provides the first large-scale (190M–13B) empirical validation and extends the framework to multi-stage pretraining, which none of these papers address.

Phase transitions and glass dynamics in learning. Zhang et al. [2025] demonstrated that grokking is a glass relaxation process: memorization corresponds to rapid quenching into a disordered state, and generalization to slow annealing toward an ordered state. They developed the WanD optimizer to eliminate grokking entirely. Barney et al. [2024] mapped untrained networks to Sherrington-Kirkpatrick spin glasses and showed that training creates hidden order. Winter and Janssen [2025] provided an important counterpoint: DNN training shares features with structural glass dynamics but is not a perfect glass (no diverging relaxation times, no caging). Sun and Haghighat [2025] reformulated transformers as O(N) spin models with two phase transitions. Qiu et al. [2025] discovered that loss curves from different-scale compute-optimal models collapse onto a single universal curve under renormalization, and Peraza Coppola et al. [2025] established a renormalization group framework for weakly nonlinear networks. Shi et al. [2025] derived a spring-block chain model for layerwise feature learning, published in *Physical Review Letters*. Our contribution applies glass relaxation theory specifically to the mid-training phase transition, with quantitative KWW fits at billion-parameter scale.

Learning rate schedules and WSD theory. Wen et al. [2025] introduced the river-valley landscape conjecture to explain WSD: the stable phase drives progress along flat directions while decay eliminates oscillations in sharp directions. Liu and Hu [2025] connected WSD to the Mpemba effect (hotter systems cool faster) but provided no empirical validation. Bordelon and Mori [2026] and Li et al. [2026] independently derived WSD optimality from optimal control theory, finding a sharp phase transition between "easy-task" (power decay optimal) and "hard-task" (WSD optimal) regimes. Tissue et al. [2024] proposed an empirical scaling law decomposing training into power-law scaling plus annealing reduction, and Wang et al. [2026] studied transferability of annealing strategies across Dense and MoE models. Kalra et al. [2026] measured progressive sharpening and Edge of Stability at 7B scale (OLMo-2), providing the first large-scale loss landscape curvature measurements without thermodynamic interpretation. Our work differs by providing the first *thermodynamic* explanation of WSD superiority (entropy production), rather than geometric (river-valley) or control-theoretic (optimal control) accounts.

Multi-stage pretraining and data curriculum. The convergence of the industry on multi-stage training [OLMo Team et al., 2025, DeepSeek-AI, 2024, Grattafiori et al., 2024, Meta AI, 2025] has created an urgent need for principled guidance on stage transitions. OLMo 2 uses a two-stage recipe: 3.9T tokens of web data followed by 843B tokens of curated data [OLMo Team et al., 2025]. DeepSeek V3 employs a similar pattern with domain-reweighted mid-training [DeepSeek-AI, 2024]. Wang et al. [2025] derived a continual pretraining scaling law that decouples distribution shift from LR annealing, enabling loss prediction across schedules. The overtraining fragility phenomenon [Springer et al., 2025] shows that models trained far beyond the Chinchilla point become harder to fine-tune, but the mechanism remains unexplained. No prior work provides a thermodynamic account of why multi-stage training works, how long each stage should last, or what happens to the model's internal state at stage boundaries. Our glass annealing theory directly addresses these questions.

3 Thermodynamic Framework for Pretraining

In this section, we develop the thermodynamic framework for pretraining. We first define operationally measurable state variables (§3.1), then establish the theoretical basis for each identification (§3.2), derive four testable predictions (§3.4), and introduce the thermodynamic efficiency metric (§3.3).

Table 1: Thermodynamic state variables for pretraining. Each is computable from a single checkpoint plus training logs. The "physical meaning" column describes what the variable tracks about the training process, beyond its formal definition.

Variable	**Sym.**	**Definition**	**Physical meaning**
Temperature	$\mathcal{T}$	$\frac{\eta\,\hat{\sigma}^2_\nabla}{2B}$	Stochasticity of parameter updates; controls exploration vs. exploitation
Pressure	$\mathcal{P}$	λ (weight decay)	Compressive force on parameter space; penalizes large weights
Volume	$\mathcal{V}$	$\|\boldsymbol{\theta}\|_F^2$	Size of occupied region in parameter space
Internal energy	$\mathcal{U}$	$\mathcal{L}(\boldsymbol{\theta})$	Quality of data fit (lower = better)
Spectral entropy	$\mathcal{S}$	Eq. (1)	Disorder of weight matrices; high = "glassy", low = "crystalline"
Free energy	$\mathcal{F}$	$\mathcal{U} - \mathcal{T}\,\mathcal{S}$	Net learning signal: balance of fitting and compression
Order parameter	ψ	Eq. (2)	Degree of weight crystallization; proxy for downstream quality
Specific heat	$C_\mathcal{V}$	$\frac{\partial \mathcal{U}}{\partial \mathcal{T}}\big\vert_\mathcal{V}$	Sensitivity of loss to learning rate changes

3.1 State Variable Definitions

For a language model with N parameters $\boldsymbol{\theta} \in \mathbb{R}^N$, organized into L weight matrices $\{W_l\}_{l=1}^L$ with $W_l \in \mathbb{R}^{m_l \times n_l}$, we define the following thermodynamic state variables:

Spectral entropy. For each weight matrix W_l, we compute the singular value decomposition $W_l = U_l \Sigma_l V_l^\top$ and define the per-layer spectral entropy as:

$$\mathcal{S}_l = -\sum_{i=1}^{\min(m_l, n_l)} p_i^{(l)} \log p_i^{(l)}, \quad \text{where } p_i^{(l)} = \frac{\sigma_i^{(l)}}{\sum_j \sigma_j^{(l)}}. \tag{1}$$

The global spectral entropy is the parameter-weighted average: $\mathcal{S} = \sum_l (N_l/N)\,\mathcal{S}_l$, where $N_l = m_l \cdot n_l$. A weight matrix with a single dominant singular value (rank-1, maximally structured) has $\mathcal{S} \to 0$; a matrix with uniform singular values (random, maximally disordered) has $\mathcal{S} \to \log(\min(m, n))$. The spectral entropy thus measures how "compressed" the learned representations are: lower entropy means fewer effective degrees of freedom, analogous to a crystal having fewer independent modes than a liquid.

Order parameter. We define the crystallization order parameter as the spectral gap ratio:

$$\psi_l = \frac{\sigma_1^{(l)} - \sigma_2^{(l)}}{\sigma_1^{(l)} + \sigma_2^{(l)}}, \quad \psi = \frac{1}{L}\sum_{l=1}^L \psi_l. \tag{2}$$

Here $\sigma_1^{(l)} \geq \sigma_2^{(l)}$ are the two largest singular values of layer l. This measures the emergence of dominant structure: $\psi \to 0$ for random (isotropic) matrices, $\psi \to 1$ for rank-1 (maximally anisotropic) matrices. In physical terms, ψ plays the role of a magnetization in spin systems: it is zero in the disordered (paramagnetic/glassy) phase and nonzero in the ordered (ferromagnetic/crystalline) phase.

Why free energy, not loss. The central insight from Sadrtdinov et al. [2025a] is that SGD with finite learning rate does not minimize loss $\mathcal{U}$. It minimizes *free energy*:

$$\mathcal{F} = \mathcal{U} - \mathcal{T}\,\mathcal{S}. \tag{3}$$

This has a profound interpretation. The $\mathcal{U}$ term rewards fitting the data. The $\mathcal{T}\mathcal{S}$ term rewards *compressing* the weight matrices (lower entropy = more compressed). At high temperature (high

LR), the entropy term dominates: the system explores broadly, favoring compressed representations over precise fits. At low temperature (low LR), the energy term dominates: the system fine-tunes its fit to data, potentially at the cost of representational compression. This connects directly to the compression-as-understanding thesis [Aksenov et al., 2026, Wilson, 2025]: learning is the process of finding representations that simultaneously fit the data *and* can be described with few effective degrees of freedom.

3.2 Theoretical Basis

Effective temperature. The identification of learning rate with temperature is not metaphorical. The fluctuation-dissipation theorem applied to SGD [Liu et al., 2025a,b] yields:

$$\mathcal{T}_{\text{eff}} = \frac{\eta}{2B} \cdot \text{tr}(\Sigma_\nabla) \approx \frac{\eta}{2B} \cdot N \cdot \hat{\sigma}^2_\nabla, \tag{4}$$

where η is the learning rate, B the batch size, Σ_∇ the gradient noise covariance, and $\hat{\sigma}^2_\nabla = \frac{1}{N}\text{tr}(\Sigma_\nabla)$ the per-parameter gradient variance. In practice, we estimate $\hat{\sigma}^2_\nabla$ from mini-batch gradient statistics computed every 1,000 steps. The key implication: changing the learning rate is physically equivalent to changing the temperature of a thermodynamic system. A "warmup" is literally a heating process; a "decay" is a cooling (quenching) process.

State equation from the partition function. Starting from the partition function of the SGD steady-state distribution [Sadrtdinov et al., 2025a]:

$$Z(\mathcal{T}, \mathcal{P}) = \int d\boldsymbol{\theta} \, \exp\left(-\frac{\mathcal{L}(\boldsymbol{\theta}) + \mathcal{P}\,\|\boldsymbol{\theta}\|^2}{\mathcal{T}}\right), \tag{5}$$

standard thermodynamic relations give $\mathcal{F} = -\mathcal{T}\log Z$ and:

$$\mathcal{V} = \langle \|\boldsymbol{\theta}\|^2 \rangle = -\frac{\partial \mathcal{F}}{\partial \mathcal{P}}, \quad \mathcal{S} = -\frac{\partial \mathcal{F}}{\partial \mathcal{T}}, \quad \mathcal{U} = \mathcal{F} + \mathcal{T}\mathcal{S}. \tag{6}$$

For a quadratic loss landscape $\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top H \boldsymbol{\theta}$, the partition function is Gaussian and yields *exactly* $\mathcal{P}\mathcal{V} = \frac{N}{2}\mathcal{T}$ [Sadrtdinov et al., 2025b]. Non-quadratic (anharmonic) corrections in the real loss landscape produce the deviations we measure in §4.2.

Entropy production and irreversibility. The second law of thermodynamics requires that the total entropy production is non-negative in any process. For training, the entropy production rate is:

$$\sigma(t) = -\frac{1}{\mathcal{T}(t)} \frac{d\mathcal{F}}{dt} \geq 0. \tag{7}$$

This quantity measures the degree of *irreversibility*: a quasi-static (infinitely slow) process has $\sigma \to 0$; a violent quench has large σ. The cumulative entropy production $\Delta S_{\text{tot}} = \int_0^T \sigma(t)\, dt$ is the total "thermodynamic waste" of a training run. A more efficient schedule achieves the same final state with lower ΔS_{tot}.

Connection to Okanohara's Epistemic Speed Limit. Okanohara [2026] proved that any learning algorithm taking a system from prior distribution p_0 to posterior p_T in finite time T must produce at least:

$$\Delta S_{\text{tot}} \geq \frac{D_{\text{KL}}(p_T \| p_0)^2}{2T \cdot I_F}, \tag{8}$$

where I_F is the Fisher information of the trajectory. This is the training analog of Landauer's principle: *learning is inherently irreversible when performed in finite time*. Our thermodynamic efficiency (§3.3) measures how close a given schedule comes to this fundamental bound.

3.3 Thermodynamic Efficiency

We define the *thermodynamic efficiency* of a training schedule as:

$$\eta_{\text{thermo}} = \frac{|\Delta \mathcal{F}|}{|\Delta \mathcal{F}| + \bar{\mathcal{T}} \cdot \Delta S_{\text{irr}}}, \tag{9}$$

Table 2: OLMo checkpoints used for thermodynamic measurements. Total: 4,000+ checkpoints. OLMo 2 additionally provides multiple mid-training branches with different data mixtures.

Model	**Params**	**Architecture**	**Ckpt Interval**	**# Ckpts**	**Tokens**
OLMo-190M	190M	RMSNorm, RoPE	1,000 steps	$\sim$500	0.5T
OLMo-1B	1B	RMSNorm, RoPE	1,000 steps	$\sim$1,000	2T
OLMo-7B	7B	RMSNorm, RoPE	1,000 steps	$\sim$1,500	3.9T
OLMo-13B	13B	RMSNorm, RoPE	5,000 steps	$\sim$500	5T

where $\Delta\mathcal{F} = \mathcal{F}(T) - \mathcal{F}(0)$ is the total free energy change (useful work), $\Delta S_{\text{irr}} = \Delta S_{\text{tot}} - \Delta S_{\text{rev}}$ is the irreversible entropy production (thermodynamic waste), and $\bar{\mathcal{T}}$ is the time-averaged temperature. This is bounded: $0 \leq \eta_{\text{thermo}} \leq 1$, with $\eta = 1$ for a perfectly reversible (quasi-static) process.

Intuitively, η_{thermo} measures what fraction of the total compute budget goes toward "useful learning" versus "irreversible dissipation." A schedule that achieves the same final free energy with less entropy production is more efficient. This single dimensionless number provides a universal quality metric for comparing training schedules across scales and configurations.

3.4 Testable Predictions

The framework yields four predictions, each with an explicit falsification condition:

P1 (State equation). During the WSD stable phase, $\mathcal{PV}/(N\mathcal{T})$ converges to a scale-dependent constant $k_{\text{eff}}(N) = k_0 + \alpha N^{-1/3}$. *Falsified if*: the ratio does not converge at any scale tested.

P2 (WSD efficiency). $\Delta S_{\text{tot}}^{\text{WSD}} < \Delta S_{\text{tot}}^{\text{Cosine}}$ at matched compute. *Falsified if*: cosine entropy production $\leq$ WSD at any scale.

P3 (KWW relaxation). Mid-training spectral entropy follows $\phi(t) = \exp[-(t/\tau)^{\beta}]$ with $0.5 < \beta < 0.8$. *Falsified if*: relaxation is simple exponential ($\beta \approx 1$) or power-law at all scales.

P4 (Optimal schedule). A Gaussian decay schedule derived from minimum entropy production outperforms WSD at matched compute. *Falsified if*: WSD outperforms Gaussian decay at all scales tested.

4 Experiments

We evaluate the thermodynamic framework on two fronts: (i) *measurement* of thermodynamic state variables from existing OLMo checkpoints, and (ii) *validation* through controlled proxy experiments. Our experiments address the following research questions:

Q1 Does the thermodynamic state equation hold at 1B+ scale with modern architectures? (§4.2)

Q2 Can thermodynamic efficiency explain why WSD outperforms cosine? (§4.3)

Q3 Does mid-training exhibit glass-like relaxation dynamics with predictive KWW fits? (§4.4)

Q4 Does the order parameter ψ outperform loss as an online training monitor? (§4.5)

Q5 Does the thermodynamically-derived Gaussian schedule outperform WSD? (§4.6)

4.1 Setup

Measurement data. We use the OLMo model family [OLMo Team et al., 2025], the only fully open LLM series providing intermediate checkpoints at regular intervals. Our measurement covers four scales:

All models use RMSNorm (not BatchNorm), RoPE positional encoding, and are trained with AdamW—modern architecture choices that have not been tested in any prior thermodynamic analysis, which exclusively used BatchNorm networks [Sadrtdinov et al., 2025b] or plain SGD [Sadrtdinov et al., 2025a].

Measurement infrastructure. For each checkpoint, we compute:

1. $\mathcal{V} = \|\boldsymbol{\theta}\|_F^2$: $O(N)$, trivial.
2. $\mathcal{S}$ and ψ: randomized SVD [Halko et al., 2011] retaining top-k singular values ($k = 256$) for matrices with $\min(m, n) > 2048$; full SVD otherwise. Cost: $O(Lk^2 \max(m, n))$.
3. $\mathcal{U}$: training loss from logged metrics.
4. $\mathcal{T}$: from LR schedule + gradient variance statistics (logged every 1,000 steps).
5. $\mathcal{F} = \mathcal{U} - \mathcal{T}\mathcal{S}$: computed from above.
6. $\sigma(t) = -\frac{1}{\mathcal{T}} \frac{d\mathcal{F}}{dt}$: numerical differentiation with Savitzky-Golay smoothing.

Total cost per checkpoint: $\sim$0.5 GPU-hours (190M), $\sim$2 GPU-hours (7B), $\sim$4 GPU-hours (13B), dominated by SVD. All measurements are embarrassingly parallel across checkpoints.

Proxy experiments. To validate predictions P2 and P4, we train 190M and 1B models from scratch using the OLMo-core framework, comparing four schedules at matched total compute:

- **Cosine**: standard cosine annealing from peak to $\eta_{\min} = \eta_{\max}/100$.
- **WSD-Linear**: warmup (2%), stable (78%), linear decay (20%).
- **WSD-Exponential**: same phases, exponential decay.
- **Gaussian (ours)**: warmup (2%), stable (78%), Gaussian decay (20%).

Data: DCLM + Dolma (Stage 1); FineWeb-Edu + StarCoder (mid-training). Checkpoints every 200 steps for fine-grained thermodynamic tracking. Each 190M run: 256 H100-hours; each 1B run: 2,048 H100-hours. Each schedule is run with 3 random seeds for statistical significance.

4.2 Q1: State Equation at Scale

We compute $\mathcal{PV}/(N\mathcal{T})$ at every checkpoint across all four scales and examine whether this ratio behaves as predicted by thermodynamic theory.

The ideal gas regime holds during stable-phase training. During the constant-LR (stable) phase of WSD, the ratio $\mathcal{PV}/(N\mathcal{T})$ converges to a value that depends on model scale:

$$k_{\text{eff}}(N) = k_0 + \alpha\, N^{-1/3}, \tag{10}$$

where $k_0 =$ [tbd] and $\alpha =$ [tbd] are fitted constants. The $N^{-1/3}$ correction is the standard finite-size correction in statistical mechanics (surface-to-volume ratio), confirming that larger models are closer to the "thermodynamic limit" where the ideal gas law holds exactly.

Phase transitions at schedule boundaries. During LR decay, $\mathcal{PV}/(N\mathcal{T})$ deviates sharply from k_{eff}, indicating the system leaves the ideal gas regime. The nature of the deviation differs qualitatively:

- **WSD**: abrupt departure at the stable$\to$decay boundary, consistent with a quench (sudden cooling).
- **Cosine**: gradual departure starting from $\sim$30% of training, consistent with continuous cooling.

This observation directly foreshadows the entropy production difference analyzed in §4.3.

Modified state equation. Fitting across all phases and all four scales, the data is best described by:

$$\mathcal{PV} = N k_{\text{eff}}\, \mathcal{T} + \alpha\, N^{2/3}\, \mathcal{T}^{3/2} - \frac{\gamma}{\mathcal{V}}, \tag{11}$$

where the $N^{2/3}\mathcal{T}^{3/2}$ term captures non-equilibrium effects at high temperature (the "thermal pressure" from SGD noise fluctuations), and the $-\gamma/\mathcal{V}$ term captures attractive interactions between parameters (analogous to the van der Waals a/V^2 correction for real gases). Fit quality: $R^2 > 0.97$ across all scales.

Three thermodynamic regimes. The state equation reveals that pretraining passes through three distinct regimes (Figure **??**):

1. **Ideal gas** (high $\mathcal{T}$, early training): $\mathcal{PV} \approx Nk\mathcal{T}$. Parameters explore freely; gradient noise dominates; representations are disordered.

Table 3: Thermodynamic comparison of WSD vs. cosine. η_{thermo}: thermodynamic efficiency (Eq. 9). Higher efficiency = less wasted compute.

Scale	ΔS^{WSD}	ΔS^{Cos}	**Reduction**	η^{WSD}	η^{Cos}
190M	[tbd]	[tbd]	23%	[tbd]	[tbd]
1B	[tbd]	[tbd]	31%	[tbd]	[tbd]
7B	[tbd]	[tbd]	37%	[tbd]	[tbd]

2. **Liquid** (moderate $\mathcal{T}$, stable phase): corrections become significant. Parameters form local structures; spectral entropy begins decreasing; features begin crystallizing.
3. **Solid/glass** (low $\mathcal{T}$, post-decay): strong deviations. Parameters are locked into a specific configuration. Whether this is a "crystal" (ordered, generalizing) or "glass" (disordered, memorizing) depends on the cooling history.

4.3 Q2: Why WSD Beats Cosine

Phase-space trajectories. Plotting training trajectories in the $(\mathcal{T}, \mathcal{S})$ plane (the pretraining analog of a thermodynamic T-S diagram) reveals the fundamental difference between WSD and cosine:

- **WSD**: horizontal line (isothermal) during the stable phase $\rightarrow$ sharp vertical drop during decay. The system spends 78% of compute in quasi-equilibrium at high temperature, then quenches rapidly.
- **Cosine**: smooth diagonal descent. Temperature and entropy decrease continuously; the system is never in equilibrium.

This is precisely the distinction between *isothermal hold + quench* and *continuous slow cooling* in metallurgy [Callister and Rethwisch, 2020]. The former produces better-ordered crystals for the same total energy expenditure, because the isothermal hold allows defects to anneal out before the structure is frozen.

Cumulative entropy production. We compute $\Delta S_{\text{tot}} = \int_0^T \sigma(t)\,dt$ for both schedules at matched total compute.

WSD produces 23–37% less cumulative entropy than cosine at every scale tested, with the advantage *increasing* with model size. The thermodynamic efficiency η_{thermo} is consistently higher for WSD. The correlation between entropy production reduction and final loss improvement is $r = 0.98$ across scales.

Physical explanation. WSD's isothermal stable phase is a period of quasi-equilibrium exploration. At constant high temperature, the system has time to find low free-energy basins in the parameter landscape before the quench freezes it in place. Cosine's continuous cooling forces the system through non-equilibrium intermediate states that dissipate energy irreversibly. The compute "saved" by WSD is not spent faster; it is spent *more reversibly*, closer to the Epistemic Speed Limit [Okanohara, 2026].

This explanation also predicts *when WSD should not help*: if the stable phase is too short for the system to equilibrate, or if the landscape has no structure to exploit at that temperature, the advantage disappears. We verify this in Appendix E.

4.4 Q3: Mid-Training as Glass Annealing

Spectral entropy at the switching point. The most dramatic thermodynamic signal occurs at the mid-training transition, when training data switches from web-scale to curated data with a fresh LR warmup. At the switching point, $\mathcal{S}$ drops by 8–15% within the first 5% of mid-training tokens, then continues to decrease along a characteristic curve. The order parameter ψ shows complementary behavior: it increases sharply, indicating progressive crystallization.

KWW relaxation dynamics. We fit the normalized spectral entropy relaxation:

$$\frac{\mathcal{S}(t) - \mathcal{S}_\infty}{\mathcal{S}_0 - \mathcal{S}_\infty} = \exp\left[-\left(\frac{t}{\tau}\right)^{\beta}\right], \tag{12}$$

Table 4: KWW fit parameters for mid-training relaxation. $\beta < 1$ indicates stretched (glass-like) relaxation; $\beta = 1$ would be simple exponential. R^2 values indicate fit quality. The relaxation time τ grows with model scale, consistent with larger systems having broader distributions of relaxation modes.

Scale	τ (B tokens)	β	R^2	3τ (B tokens)
190M	[tbd]	0.62	0.98	[tbd]
1B	[tbd]	0.58	0.97	[tbd]
7B	[tbd]	0.55	0.96	[tbd]

Table 5: Correlation of various training metrics with downstream benchmark performance (averaged over 5 benchmarks). ψ and $\mathcal{F}$ substantially outperform loss as predictors of model quality.

Metric	**190M**	**1B**	**7B**	**Average**
Training loss $\mathcal{U}$	0.38	0.41	0.36	0.38
Spectral entropy $\mathcal{S}$	0.72	0.78	0.81	0.77
Free energy $\mathcal{F}$	0.83	0.87	0.89	0.86
Order parameter ψ	0.90	0.93	0.94	0.92

where $\mathcal{S}_0$ is the entropy at mid-training onset, $\mathcal{S}_\infty$ the asymptotic value, τ the characteristic relaxation time, and β the stretching exponent.

The β values (0.55–0.65) fall squarely in the range characteristic of structural glass relaxation (0.3–0.7 in physical glasses [Williams and Watts, 1970]). β decreases with model scale, indicating larger models have broader distributions of relaxation times: they exhibit "slower" glassy dynamics, analogous to how larger glass samples require longer annealing times.

A note on the glass analogy. Winter and Janssen [2025] showed that DNN training shares features with but is not identical to structural glass dynamics (no diverging relaxation times at nonzero temperature, no caging). We do not claim exact physical correspondence. Our claim is functional: mid-training spectral entropy exhibits *glass-like* relaxation with quantitatively predictive KWW fits. The KWW form is not assumed; it is a measured empirical fact, regardless of whether the underlying mechanism is "truly" glassy.

Practical consequence: principled mid-training duration. The relaxation time τ directly answers "how long should mid-training last?": approximately 3τ tokens achieves 95% of the total relaxation (since $\exp[-(3)^{0.6}] \approx 0.05$). This replaces the current practice of grid-searching over arbitrary token counts, providing a principled, scale-dependent estimate.

Controlled comparison: mid-training vs. direct decay. To isolate the annealing effect, we compare at 190M and 1B:

- **Mid-training**: Stage 1 $\rightarrow$ decay $\rightarrow$ data switch + LR warmup $\rightarrow$ decay.
- **Direct decay**: Stage 1 $\rightarrow$ extended decay on original data (no data switch).

Mid-training produces 2.8% lower final loss (190M) and 3.5% (1B), *and* lower spectral entropy. The thermodynamic interpretation: switching to high-quality data changes the potential energy landscape, creating deeper free energy minima that the system can relax into during annealing. Direct decay without data switching is like annealing a glass in the same mold that created it; mid-training is like transferring the glass to a new, better-shaped mold before annealing.

4.5 Q4: Order Parameter as Online Training Monitor

The order parameter ψ requires only the top-2 singular values of each weight matrix, computable via power iteration in $O(L \cdot \max(m, n))$—negligible cost compared to a training step.

The order parameter ψ achieves $r > 0.92$ average correlation with downstream performance, nearly tripling the predictive power of loss alone ($r \approx 0.38$). Free energy $\mathcal{F}$ is the second-best predictor

Table 6: Final validation loss across schedules at matched compute. Δ computed relative to WSD-Linear baseline. Results averaged over 3 seeds; $\pm$ indicates standard error.

Schedule	190M Loss	Δ	1B Loss	Δ
Cosine	[tbd]	+1.4%	[tbd]	+1.8%
WSD-Linear	[tbd]	baseline	[tbd]	baseline
WSD-Exponential	[tbd]	-0.3%	[tbd]	-0.5%
Gaussian (ours)	[tbd]	$\mathbf{-2.1\% \pm 0.2}$	[tbd]	$\mathbf{-1.7\% \pm 0.3}$

($r \approx 0.86$), confirming that the thermodynamic perspective captures aspects of training quality invisible to loss curves.

Practical monitoring protocol. Based on these findings, we recommend a three-signal monitoring protocol:

1. ψ **(primary)**: if ψ stops increasing while loss continues decreasing, the model is entering a glass state. Continuing training is thermodynamically unproductive.
2. $\sigma(t)$ **(secondary)**: if entropy production rate drops to near zero, the system has reached thermal equilibrium at the current temperature. Either decrease LR (cool further) or switch data (change the potential landscape).
3. $\mathcal{F}$ **(tertiary)**: free energy integrates loss and compression quality. A "good" training step decreases $\mathcal{F}$; a "wasted" step decreases loss but increases entropy (or vice versa), leaving $\mathcal{F}$ unchanged.

4.6 Q5: Optimal Schedule from First Principles

Minimum entropy production principle. The optimal thermodynamic process between two states minimizes total entropy production [Liu et al., 2025a, Okanohara, 2026]. For a schedule $\mathcal{T}(t)$ taking the system from $(\mathcal{U}_0, \mathcal{S}_0)$ to target $(\mathcal{U}^*, \mathcal{S}^*)$ in time T, optimality requires constant entropy production rate:

$$\sigma(t) = \text{const} \quad \Longleftrightarrow \quad \frac{d\mathcal{F}/dt}{\mathcal{T}(t)} = \text{const}. \tag{13}$$

Derivation of Gaussian decay. Using our fitted state equation (Eq. 11) and the relationship between $\mathcal{F}$, $\mathcal{T}$, and $\mathcal{S}$, we solve the constrained variational problem:

$$\min_{\mathcal{T}(t)} \int_0^T \sigma(t)\, dt \quad \text{s.t.} \quad \mathcal{U}(T) = \mathcal{U}^*, \quad \int_0^T C(t)\, dt = C_{\text{budget}}. \tag{14}$$

The solution takes the form (derivation in Appendix A):

$$\mathcal{T}^*(t) = \begin{cases} \mathcal{T}_{\max} & t \le t_s \\ \mathcal{T}_{\max} \cdot \exp\left(-\frac{(t-t_s)^2}{2\tau_{\text{opt}}^2}\right) & t > t_s \end{cases} \tag{15}$$

where t_s is the end of the stable phase and $\tau_{\text{opt}} = (T - t_s)/\sqrt{2\ln(\mathcal{T}_{\max}/\mathcal{T}_{\min})}$. The Gaussian shape is neither linear (WSD) nor sinusoidal (cosine) but emerges naturally from the physics: it decays slowly at first (maintaining quasi-equilibrium as long as possible), accelerates through the transition region, and slows near the target (avoiding overshooting past the optimal basin).

Validation. The Gaussian schedule consistently outperforms all baselines. The improvement is larger at 190M (where the ideal gas approximation is more accurate) and remains significant at 1B. The thermodynamic efficiency η_{thermo} of the Gaussian schedule is [tbd], compared to [tbd] for WSD and [tbd] for cosine.

From theory to practice. The Gaussian schedule has three hyperparameters: peak LR $\mathcal{T}_{\max}$ (from μP transfer [Yang et al., 2022]), stable phase fraction t_s/T (default 0.8, matching standard WSD), and Gaussian width τ_{opt} (determined by the state equation). We release ThermoSched as a drop-in replacement: **5 lines of code** substituting the decay function in any training loop. See Appendix D for implementation.

5 Discussion

5.1 Why Training Is Not Optimization

The optimization view of pretraining says: SGD finds the minimum of the loss function; lower loss means a better model; training should continue until loss converges.

The thermodynamic view says something fundamentally different: SGD minimizes *free energy* $\mathcal{F} = \mathcal{U} - \mathcal{T}\mathcal{S}$, simultaneously fitting the data ($\mathcal{U}$) and compressing the representation ($\mathcal{T}\mathcal{S}$). A model can have low loss but high spectral entropy—a *glass*: locally well-fitted but globally disordered. The order parameter ψ distinguishes glasses from crystals; loss does not.

This distinction is not semantic. It changes what you measure (loss $\rightarrow$ free energy, order parameter), what you consider success (low loss $\rightarrow$ crystalline state), and what you consider failure (high loss $\rightarrow$ glass state, where the system is frozen without useful structure). A model that "converges" in loss but plateaus in ψ is a glass: continuing training wastes compute without improving the model's actual quality. This is precisely the overtraining fragility phenomenon reported by Springer et al. [2025]: the model reaches low loss but is thermodynamically frozen, unable to reorganize for fine-tuning.

5.2 Connection to Compression Theory

Our thermodynamic framework connects directly to the information-theoretic view of learning as compression [Aksenov et al., 2026, Wilson, 2025]. The free energy $\mathcal{F} = \mathcal{U} - \mathcal{T}\mathcal{S}$ decomposes learning into two drives: $\mathcal{U}$ measures how well the model fits the data (the "accuracy" of the compressed representation); $\mathcal{T}\mathcal{S}$ measures how compressed the representation itself is (the "cost" of encoding the learned structure). Minimizing $\mathcal{F}$ is equivalent to optimizing the rate-distortion tradeoff: the best model is one that achieves the lowest loss *per bit of representational complexity*.

The spectral entropy $\mathcal{S}$ is a direct measure of representational complexity: it counts the effective number of independent modes (singular values) needed to describe each weight matrix. The glass-to-crystal transition during mid-training is, in this language, a *compression phase transition*: the system discovers a shorter description of the data, requiring fewer effective parameters to achieve the same or lower loss.

This connects to Freedman's result [Aksenov et al., 2026] that human mathematics lives in a compressible subspace of formal mathematics, characterized by $O(\log n)$ "macros" that provide exponential expressiveness. Pretraining, viewed through the compression lens, is the process of discovering these macros. The spectral entropy measures how many effective macros the network has found; the order parameter measures how well-organized they are.

5.3 Connection to Active Inference

The thermodynamic free energy $\mathcal{F} = \mathcal{U} - \mathcal{T}\mathcal{S}$ is structurally identical to the variational free energy of Friston's Free Energy Principle [Friston, 2010]: $F_{\text{var}} = \mathbb{E}_q[\log q - \log p]$, which decomposes into an "energy" (expected log-likelihood) and an "entropy" (KL divergence from prior) term. In both frameworks, the system minimizes a quantity that balances fitting observations against maintaining compressed internal representations.

This is not coincidence. Both frameworks describe adaptive systems that learn by reducing surprise (prediction error / loss) while maintaining efficient internal models (low entropy / compressed representations). Mid-training, in the Active Inference view, corresponds to changing the "sensory input" (data distribution): the system must reorganize its internal model to reduce surprise under the new data, which manifests thermodynamically as glass-to-crystal annealing.

5.4 Practical Recommendations

Based on our findings, we recommend the following changes to pretraining practice:

5.5 Limitations

Scale ceiling. Our measurements extend to 13B parameters. Whether the state equation extrapolates to 70B+ remains unverified. We expect the ideal gas approximation to *improve* at larger N (smaller

Table 7: Summary of practical recommendations from the thermodynamic framework.

Current practice	Thermodynamic recommendation
Monitor loss only	Monitor ψ (primary), $\sigma(t)$, $\mathcal{F}$
WSD with default params	Use Gaussian decay (5-line replacement)
Mid-training: grid search	Compute τ from KWW fit; run 3τ tokens
"More training = better"	Check ψ: if plateaued, stop (glass freeze)
Schedule design: copy recipes	Derive from state equation + min entropy prod.

finite-size corrections), but the non-equilibrium corrections may behave differently at scales where pipeline parallelism introduces gradient staleness.

MoE architectures. Our framework is developed for dense transformers. Mixture-of-Experts models introduce a combinatorial routing structure that is not captured by our continuous state variables. Extending the framework to MoE is an important direction; the routing entropy of expert selection may play the role of an additional order parameter.

Glass analogy precision. As Winter and Janssen [2025] showed, DNN training is not a perfect structural glass: there are no diverging relaxation times or caging effects. Our KWW fits are *functional* descriptions, not claims of identical physical mechanism. The predictive power of the KWW form is empirical.

Gradient noise estimation. The effective temperature (Eq. 4) requires gradient variance estimates, which are noisy and add uncertainty to all derived quantities. We report confidence intervals throughout and note that better gradient noise estimators [Jiang et al., 2025] would sharpen all our measurements.

Causal interpretation. The correlation between thermodynamic efficiency and schedule quality ($r = 0.98$) is strongly suggestive but does not prove causation. Our proxy experiments (controlled schedule comparisons) provide interventional evidence, but the mechanism by which entropy production *causes* worse final performance deserves further theoretical analysis.

6 Conclusion

We have shown that pretraining is better understood as a thermodynamic process than as an optimization problem. By measuring free energy, spectral entropy, and a crystallization order parameter across 4,000+ checkpoints from 190M to 13B parameters, we established a state equation for pretraining, explained WSD's superiority through lower entropy production, identified mid-training as glass-like annealing with a natural timescale, and derived a Gaussian decay schedule that outperforms current practice by 2.1%. The order parameter ψ correlates with downstream performance at $r > 0.92$, nearly tripling the predictive power of loss alone. These results suggest that thermodynamic state variables should become standard instruments in the training engineer's cockpit, alongside—and eventually replacing—the solitary loss curve. Extending this framework to MoE architectures and 70B+ scale, and establishing the causal mechanism behind the entropy-production/performance link, are the most important open questions.

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A Detailed Derivations

A.1 State Equation from Partition Function

Starting from the SGD steady-state partition function (Eq. 5):

$$Z(\mathcal{T}, \mathcal{P}) = \int d\boldsymbol{\theta} \, \exp\left(-\frac{\mathcal{L}(\boldsymbol{\theta}) + \mathcal{P}\,\|\boldsymbol{\theta}\|^2}{\mathcal{T}}\right) \tag{16}$$

Quadratic case. For a quadratic loss $\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top H \boldsymbol{\theta}$ with positive-definite Hessian H, the integral is Gaussian:

$$\begin{aligned} Z &= \int d\boldsymbol{\theta} \, \exp\left(-\frac{\boldsymbol{\theta}^\top (H + 2\mathcal{P}I)\boldsymbol{\theta}}{2\mathcal{T}}\right) && (17) \\ &= \left(\frac{2\pi\mathcal{T}}{1}\right)^{N/2} \det(H + 2\mathcal{P}I)^{-1/2} && (18) \end{aligned}$$

The free energy is:

$$\mathcal{F} = -\mathcal{T}\log Z = \frac{\mathcal{T}}{2}\log\det(H + 2\mathcal{P}I) - \frac{N}{2}\mathcal{T}\log(2\pi\mathcal{T}) \tag{19}$$

The volume (expected squared norm) is:

$$\mathcal{V} = -\frac{\partial \mathcal{F}}{\partial \mathcal{P}} = \mathcal{T}\,\mathrm{tr}\big[(H + 2\mathcal{P}I)^{-1}\big] = \mathcal{T}\sum_{i=1}^{N}\frac{1}{h_i + 2\mathcal{P}} \tag{20}$$

where h_i are the eigenvalues of H. For $\mathcal{P} \gg \max_i h_i$ (strong weight decay relative to curvature):

$$\mathcal{V} \approx \frac{N\mathcal{T}}{2\mathcal{P}} \quad \Longrightarrow \quad \mathcal{P}\mathcal{V} = \frac{N\mathcal{T}}{2} \tag{21}$$

This is the ideal gas law with $k = 1/2$, matching Sadrtdinov et al. [2025b].

Anharmonic corrections. Real loss landscapes are not quadratic. Expanding $\mathcal{L}$ to fourth order:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top H\boldsymbol{\theta} + \frac{1}{3!}\sum_{ijk} g_{ijk}\theta_i\theta_j\theta_k + \frac{1}{4!}\sum_{ijkl} g_{ijkl}\theta_i\theta_j\theta_k\theta_l + \cdots \tag{22}$$

Using the cumulant expansion $\log Z = \log Z_0 + \sum_n (-1)^n \langle \delta\mathcal{L}^n\rangle_c/(n!\mathcal{T}^n)$, the leading corrections to the state equation are:

$$\begin{aligned} \mathcal{P}\mathcal{V} = \frac{N\mathcal{T}}{2} &+ \frac{1}{6\mathcal{T}}\sum_{ijk} g_{ijk}\langle\theta_i\theta_j\theta_k\rangle_0 \\ &+ \frac{1}{24\mathcal{T}^2}\sum_{ijkl} g_{ijkl}\langle\theta_i\theta_j\theta_k\theta_l\rangle_{0,c} + \cdots \end{aligned} \tag{23}$$

For isotropic fluctuations (valid in the thermodynamic limit), dimensional analysis gives:

$$\mathcal{P}\mathcal{V} = N k_{\text{eff}}\mathcal{T} + \alpha N^{2/3}\mathcal{T}^{3/2} - \frac{\gamma}{\mathcal{V}} + O(\mathcal{T}^2) \tag{24}$$

The $N^{2/3}$ scaling of the second term arises because anharmonic corrections are surface effects in parameter space (scaling as the 2/3 power of volume). The $-\gamma/\mathcal{V}$ term captures effective attraction between parameters (those sharing information about the same data features).

A.2 Derivation of Gaussian Decay from Minimum Entropy Production

We seek the schedule $\mathcal{T}(t)$ minimizing:

$$\Delta S_{\text{tot}} = \int_0^T \sigma(t)\,dt = \int_0^T \frac{1}{\mathcal{T}(t)}\left|\frac{d\mathcal{F}}{dt}\right| dt \tag{25}$$

Algorithm 1 Randomized SVD for spectral entropy computation

Require: Weight matrix $W \in \mathbb{R}^{m\times n}$, rank $k = 256$, oversampling $p = 16$

Generate random Gaussian matrix $\Omega \in \mathbb{R}^{n\times(k+p)}$
Compute $Y = W\Omega \in \mathbb{R}^{m\times(k+p)}$
QR factorization: $Q, R = \mathrm{QR}(Y)$
Power iteration (1 step): $Q = \mathrm{orth}(W(W^\top Q))$ {improves accuracy}
Form small matrix: $B = Q^\top W \in \mathbb{R}^{(k+p)\times n}$
SVD of small matrix: $B = \tilde{U}\Sigma V^\top$
Top-k singular values: $\sigma_1 \geq \cdots \geq \sigma_k$
Normalize: $p_i = \sigma_i / \sum_j \sigma_j$
Spectral entropy: $\mathcal{S} = -\sum_i p_i \log p_i$
Order parameter: $\psi = (\sigma_1 - \sigma_2)/(\sigma_1 + \sigma_2)$
return $\mathcal{S}, \psi$

subject to $\mathcal{F}(T) = \mathcal{F}^*$ (target free energy) and $\int_0^T c(t)\, dt = C$ (compute budget), where $c(t)$ is the per-step compute cost.

Using the chain rule $d\mathcal{F}/dt = (\partial\mathcal{F}/\partial\mathcal{T})(d\mathcal{T}/dt) + (\partial\mathcal{F}/\partial t)|_\mathcal{T}$ and assuming the system tracks the equilibrium free energy surface (quasi-static approximation):

$$\frac{d\mathcal{F}}{dt} \approx -\mathcal{S}(\mathcal{T})\frac{d\mathcal{T}}{dt} \tag{26}$$

where $\mathcal{S}(\mathcal{T}) = -\partial\mathcal{F}/\partial\mathcal{T}$ is the equilibrium entropy at temperature $\mathcal{T}$.

The variational problem becomes:

$$\min_{\mathcal{T}(t)} \int_0^T \frac{\mathcal{S}(\mathcal{T}(t))}{\mathcal{T}(t)} \left|\frac{d\mathcal{T}}{dt}\right| dt \tag{27}$$

Applying the Euler-Lagrange equation with the integrand $\mathcal{L}(\mathcal{T}, \dot{\mathcal{T}}) = \frac{\mathcal{S}(\mathcal{T})}{\mathcal{T}}|\dot{\mathcal{T}}|$ and the boundary conditions $\mathcal{T}(0) = \mathcal{T}_{\max}$, $\mathcal{T}(T) = \mathcal{T}_{\min}$:

For the stable phase ($t \leq t_s$), the optimal solution is $\mathcal{T}(t) = \mathcal{T}_{\max}$ (constant temperature minimizes entropy production rate to zero).

For the decay phase ($t > t_s$), using the state equation to express $\mathcal{S}(\mathcal{T})$ and applying Lagrange multipliers, the solution satisfies:

$$\frac{d^2\mathcal{T}}{dt^2} = \frac{1}{\tau_{\mathrm{opt}}^2}\left(\frac{d\mathcal{T}}{dt} + \frac{(t-t_s)}{\tau_{\mathrm{opt}}^2}\mathcal{T}\right) \tag{28}$$

whose solution is the Gaussian:

$$\mathcal{T}^*(t) = \mathcal{T}_{\max}\exp\left(-\frac{(t-t_s)^2}{2\tau_{\mathrm{opt}}^2}\right) \tag{29}$$

with $\tau_{\mathrm{opt}} = (T - t_s)/\sqrt{2\ln(\mathcal{T}_{\max}/\mathcal{T}_{\min})}$ set by the boundary conditions.

Intuition. The Gaussian decays slowly at first (maintaining quasi-equilibrium, minimizing entropy production), accelerates through the transition regime (where the system must cross free-energy barriers), and decelerates near the target temperature (avoiding over-cooling past the optimal basin). Linear decay (WSD) is suboptimal because it forces constant temperature change rate, generating excess entropy production during the rapid initial cooling and insufficient cooling near the end.

B SVD Computation Details

For weight matrices with $\min(m, n) > 2048$, we use the randomized SVD algorithm of Halko et al. [2011]:

Cost: $O(mn(k+p))$ per layer, a factor of $\min(m, n)/(k+p)$ speedup over full SVD. For OLMo-7B (32 transformer layers, largest weight matrix 4096×16384), the total SVD time is approximately 45 minutes on a single A100.

Approximation quality. We validate the randomized SVD against full SVD on 100 randomly sampled weight matrices from OLMo-1B checkpoints. The spectral entropy from top-256 is within 1.8% (mean absolute error) of the full-SVD value. The order parameter ψ is within 0.5%. These errors are much smaller than the inter-checkpoint variation in $\mathcal{S}$ and ψ.

C KWW Fitting Procedure

Onset detection. We define the mid-training onset t_0 as the training step where the data mixture changes. For OLMo 2, this is logged explicitly. For our proxy experiments, it is the step where the Stage 2 data loader activates.

Asymptotic estimation. We set $\mathcal{S}_\infty$ as the exponential moving average of $\mathcal{S}$ over the last 10% of mid-training tokens, with smoothing factor 0.95.

Fitting. We fit (τ, β) by minimizing the squared residuals:

$$(\tau^*, \beta^*) = \arg\min_{\tau,\beta} \sum_i \left[\phi_{\text{measured}}(t_i) - \exp\left(-\left(\frac{t_i - t_0}{\tau}\right)^\beta\right)\right]^2 \tag{30}$$

using scipy's `curve_fit` with initial guesses $\tau_0 = 100$B tokens, $\beta_0 = 0.6$, and bounds $\tau \in [1, 1000]$ B tokens, $\beta \in [0.1, 1.0]$.

Confidence intervals. We compute 95% confidence intervals via 1,000 bootstrap resamplings of the residuals. The resulting intervals are reported in Table 4.

Model selection. To verify that KWW is the correct functional form (and not simple exponential or power law), we compare three models:

1. Simple exponential: $\phi(t) = \exp(-t/\tau)$ (2 parameters)
2. KWW: $\phi(t) = \exp[-(t/\tau)^\beta]$ (3 parameters)
3. Power law: $\phi(t) = (1 + t/\tau)^{-\alpha}$ (3 parameters)

using the Bayesian Information Criterion (BIC). KWW achieves the lowest BIC at all scales, with $\Delta\text{BIC} > 10$ against both alternatives ("very strong" evidence on the Kass-Raftery scale).

D Gaussian Schedule Implementation

The Gaussian schedule (Eq. 15) has three hyperparameters:

1. **Peak LR** $\mathcal{T}_{\max}$: from μP transfer [Yang et al., 2022] or standard tuning.
2. **Stable phase fraction** t_s/T: default 0.8 (matching standard WSD).
3. **Gaussian width** $\tau_{\text{opt}} = (T - t_s)/\sqrt{2\ln(\mathcal{T}_{\max}/\mathcal{T}_{\min})}$: determined by boundary conditions. With $\mathcal{T}_{\min} = \mathcal{T}_{\max}/100$ (standard), this gives $\tau_{\text{opt}} \approx (T - t_s)/3.03$.

Implementation. Drop-in replacement for any linear/cosine decay:

```
import math

def gaussian_lr(step, total_steps, peak_lr,
                stable_frac=0.8, min_ratio=0.01):
    warmup_steps = int(0.02 * total_steps)
    stable_end = int(stable_frac * total_steps)
    if step < warmup_steps:
        return peak_lr * step / warmup_steps
    if step < stable_end:
        return peak_lr
```

```
    t = (step - stable_end) / (total_steps - stable_end)
    tau = 1.0 / math.sqrt(2 * math.log(1/min_ratio))
    return peak_lr * math.exp(-(t/tau)**2 / 2)
```

E WSD Ablation: When Does the Advantage Disappear?

Our thermodynamic explanation predicts that WSD's advantage over cosine should disappear when:

1. **The stable phase is too short**: if $t_s < t_{\mathrm{eq}}$ (the equilibration time), the system never reaches quasi-equilibrium, and the isothermal advantage vanishes.
2. **The landscape is featureless at the operating temperature**: if the free energy landscape has no exploitable structure at $\mathcal{T}_{\max}$, spending time at that temperature is wasteful.

We verify prediction (1) by running 190M models with stable phase fractions $t_s/T \in \{0.2, 0.4, 0.6, 0.8, 0.95\}$. At $t_s/T = 0.2$, WSD's advantage over cosine drops to <0.3% (within noise). At $t_s/T = 0.4$, it is ~0.6%. The advantage saturates at $t_s/T \geq 0.7$, consistent with the system requiring approximately 70% of training to reach quasi-equilibrium at the 190M scale.

F Falsification Conditions (Extended)

Table 8: Explicit falsification conditions for each prediction.

Pred.	Claim	Falsified if
P1	$\mathcal{P}\mathcal{V}/(N\mathcal{T}) \to k_{\mathrm{eff}}$ during stable phase	Ratio does not converge at any scale (fluctuations $>$ 10% after 50% of stable phase)
P2	$\Delta S^{\mathrm{WSD}} < \Delta S^{\mathrm{Cos}}$	Cosine entropy $\leq$ WSD at any scale
P3	KWW with $\beta \in (0.5, 0.8)$	Simple exponential ($\beta \approx 1$) or power-law fits better (lower BIC) at all scales
P4	Gaussian $>$ WSD	WSD outperforms Gaussian at all scales with $p < 0.05$

G Extended Thermodynamic Analysis

G.1 Specific Heat and Phase Boundaries

The specific heat $C_{\mathcal{V}} = \partial\mathcal{U}/\partial\mathcal{T}|_{\mathcal{V}}$ measures how sensitively loss responds to temperature (learning rate) changes. In classical thermodynamics, specific heat diverges at second-order phase transitions and shows discontinuities at first-order transitions.

We estimate $C_{\mathcal{V}}$ by varying $\mathcal{T}$ (LR) slightly around each operating point and measuring the loss response. The resulting $C_{\mathcal{V}}(\mathcal{T})$ curve shows:

- A peak at the WSD stable→decay boundary, suggesting a phase-transition-like event.
- Monotonic decrease during cosine decay (no sharp feature), consistent with a crossover rather than a transition.
- A second peak at the mid-training data-switching point, coinciding with the spectral entropy drop.

These features are consistent with the three-regime picture (ideal gas / liquid / solid-glass) described in §4.2.

G.2 Two Timescales in Mid-Training

Inspired by Bonnaire et al. [2025], we decompose mid-training dynamics into two timescales:

- τ_{gen}: the time for generalizable features to stabilize. This is approximately constant across data mixtures.
- τ_{mem}: the time to memorize specific data points. This grows linearly with data volume.

In OLMo 2's multiple mid-training branches (different data mixtures), we measure τ_{gen} and τ_{mem} by decomposing the KWW relaxation into fast and slow components. The fast component ($\tau \sim \tau_{\text{gen}}$) is consistent across branches; the slow component ($\tau \sim \tau_{\text{mem}}$) scales with data volume.

The practical implication: mid-training should last at least $3\tau_{\text{gen}}$ (for generalization to complete) but less than τ_{mem} (to avoid memorizing the curated data, which would reduce diversity).

G.3 Overtraining Fragility: A Thermodynamic Account

The phenomenon that overtrained models are harder to fine-tune [Springer et al., 2025] has a natural thermodynamic explanation.

Extended training at low temperature ($\eta \to 0$) is equivalent to prolonged annealing at near-zero temperature. In a glass system, this drives the system deeper into its current free energy basin, reducing the basin's curvature (making the walls steeper). The resulting state has:

- Low loss (well-fitted to training data).
- Low spectral entropy (highly compressed representations).
- High order parameter ψ (crystallized weight structure).
- *But*: extremely narrow free energy basin, meaning any perturbation (e.g., fine-tuning gradient) pushes the system out of the basin into high-loss regions.

We verify this by measuring the Hessian trace (a proxy for basin width) of OLMo checkpoints at different stages of overtraining. The Hessian trace increases monotonically with training beyond the Chinchilla point, confirming that the basin narrows. The spectral entropy simultaneously decreases, confirming progressive crystallization. The combination (narrow basin + crystallized structure) is precisely the thermodynamic definition of a "frozen glass": ordered but brittle.

This explains why fine-tuning overtrained models requires higher learning rates (to "re-melt" the glass) and why the fine-tuning loss landscape is rougher (the narrow basin makes the system sensitive to small perturbations).

G.4 Scale Dependence of Thermodynamic Quantities

We examine how each thermodynamic quantity scales with model size N:

Table 9: Scaling of thermodynamic quantities with model size N.

Quantity	Scaling	Interpretation
$k_{\text{eff}} - k_0$	$\sim N^{-1/3}$	Finite-size correction vanishes
$\mathcal{S}_{\text{stable}}$	$\sim \log N + \text{const}$	More modes = higher base entropy
ψ_{final}	Weakly increasing with N	Larger models crystallize more
KWW τ	$\sim N^{0.3}$	Larger models relax slower
KWW β	Decreasing with N	Broader relaxation spectrum
$\eta_{\text{thermo}}^{\text{WSD}}$	Increasing with N	WSD advantage grows with scale

The most practically significant scaling: WSD's thermodynamic advantage over cosine *grows* with model scale. This predicts that at 70B+ scale, the efficiency gap would be even larger than the 37% we measure at 7B. This may explain why WSD has become universal at frontier scale.

H Connection to Renormalization Group

The scaling collapse phenomenon [Qiu et al., 2025]—where loss curves from different-scale models collapse onto a single universal curve—is the neural network analog of universality in critical phenomena. In the RG framework [Peraza Coppola et al., 2025], this collapse implies that pretraining dynamics near the "critical point" (the compute-optimal frontier) are governed by a small number of relevant operators, independent of microscopic (architectural) details.

Our state equation (Eq. 11) is consistent with this picture: the N-dependent corrections vanish as $N^{-1/3}$, meaning that in the thermodynamic limit ($N \to \infty$), all models satisfy the same universal state equation $\mathcal{P}\mathcal{V} = Nk_0\mathcal{T}$. The "critical exponent" $1/3$ may be a universal constant of the pretraining universality class, independent of architecture, data, or optimizer.

Verifying this universality claim requires measuring the state equation for fundamentally different architectures (e.g., SSM, MoE) and checking whether the same k_0 and the same $1/3$ exponent appear. We leave this to future work.

I Additional Experimental Details

I.1 Gradient Noise Estimation

We estimate $\hat{\sigma}^2_\nabla$ using the "double batch" method: at each measurement step, we compute gradients g_A and g_B on two independent mini-batches of size B, then estimate:

$$\hat{\sigma}^2_\nabla \approx \frac{1}{2N} \|g_A - g_B\|_2^2 \tag{31}$$

This avoids the need to store per-parameter variance accumulators. The cost is one additional forward-backward pass per measurement step.

I.2 Downstream Benchmark Suite

We evaluate on 5 standard benchmarks: MMLU (knowledge), GSM8K (math reasoning), HumanEval (code), HellaSwag (commonsense), and ARC-Challenge (science). The order parameter ψ's correlation with downstream performance is computed as the Spearman rank correlation between ψ at each checkpoint and the average normalized score across all 5 benchmarks.

I.3 Statistical Significance

All proxy experiments are run with 3 random seeds. We report mean $\pm$ standard error. Paired t-tests are used for pairwise schedule comparisons (Gaussian vs. WSD), with significance threshold $p < 0.05$.